2412E1033

Roll No.

Course: B.Tech Computer Engineering /
B.Tech Computer Science and Engineering
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)
Subject: Design and Analysis of Algorithms
Subject Code: ETCS-207
Semester: III

Time: 03 Hours **Max Marks: 70**

Instructions to the Students:

1. This Question paper consists of two Sections. All sections are compulsory.
2. Section A comprises 10 questions of short answer type. All questions are compulsory. Each question carries 02 marks.
3. Section B comprises 8 long answer type questions out of which students must attempt any 5. Each question carries 10 marks.
4. Do not write anything on the question paper.

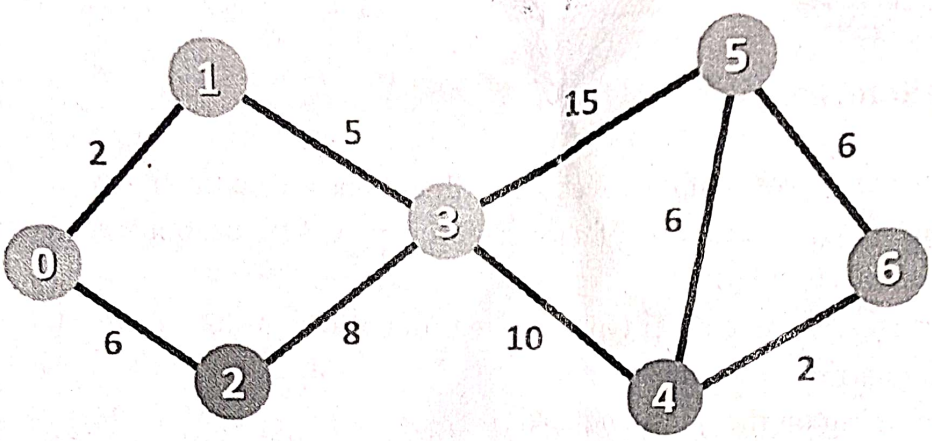
Q.No.	**SECTION –A (SHORT ANSWER TYPE QUESTIONS)**	**Marks**
1. a.	Write the time complexities of (i) Binary search (ii) Max min	(2)
b.	Define algorithm.	(2)
c.	Define order of growth.	(2)
d.	What are the types of greedy techniques?	(2)
e.	Define the terms: i.Cyclic graph ii.Degree of node	(2)
f.	Define NP hard and NP Complete problem. State example of each.	(2)
g.	Name the different graph traversal techniques	(2)
h.	Define graph colouring.	(2)
i.	State sum of subsets problem.	(2)
j.	State Cooke's theorem	(2)

SECTION –B (LONG ANSWER TYPE QUESTIONS)

2.	Explain Asymptotic Notations with example.	(10)

3. Write an algorithm to find maximum and minimum in a set of members using divide and conquer technique. Explain with an example. (10)
4. Write a recursive binary search algorithm. Explain with an example. (10)
5. a) Write merge sort algorithm to sort a set of numbers in ascending order. (10)
   b) Trace the merge sort algorithm to sort the following elements.
   41, 32, 11, 92, 66,74,87,38.
6. Find the shortest distance from node 1 to all other nodes in figure 1 using Dijikstras algorithm (10)

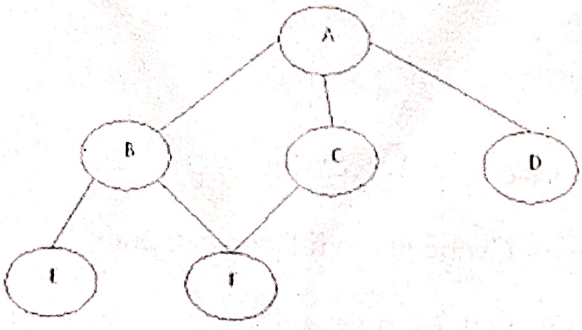
Figure 1.



7. Solve the job sequencing problem with number of jobs n=5.Their profits are (p1,p2,p3,p4,p5)=(1,5,20,15,10) and deadlines are(d1,d2,d3,d4,d5)=(1,2,4,1,3). (10)

8. a) Explain DFS algorithm. Traverse the following graph in figure 2 using DFS (10)

Figure 2.



9. For the knapsack problem ,n=3,m=20,(p1,p2,p3)=(25,24,15) and (w1,w2,w3)=(18,15,10).Find the feasible solution and optimal solution (10)

===END OF PAPER===